How to become a SOC Engineer

The silent heroes

Presented by Nhu Pham

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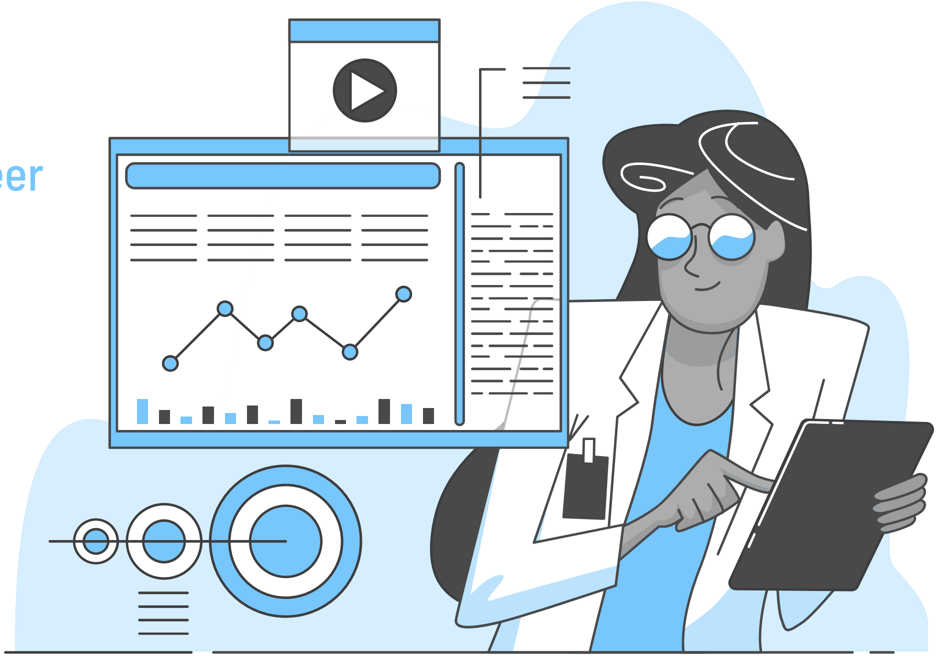


About me

PHẠM HOÀNG NHU
6 years of experience in Cyber Security

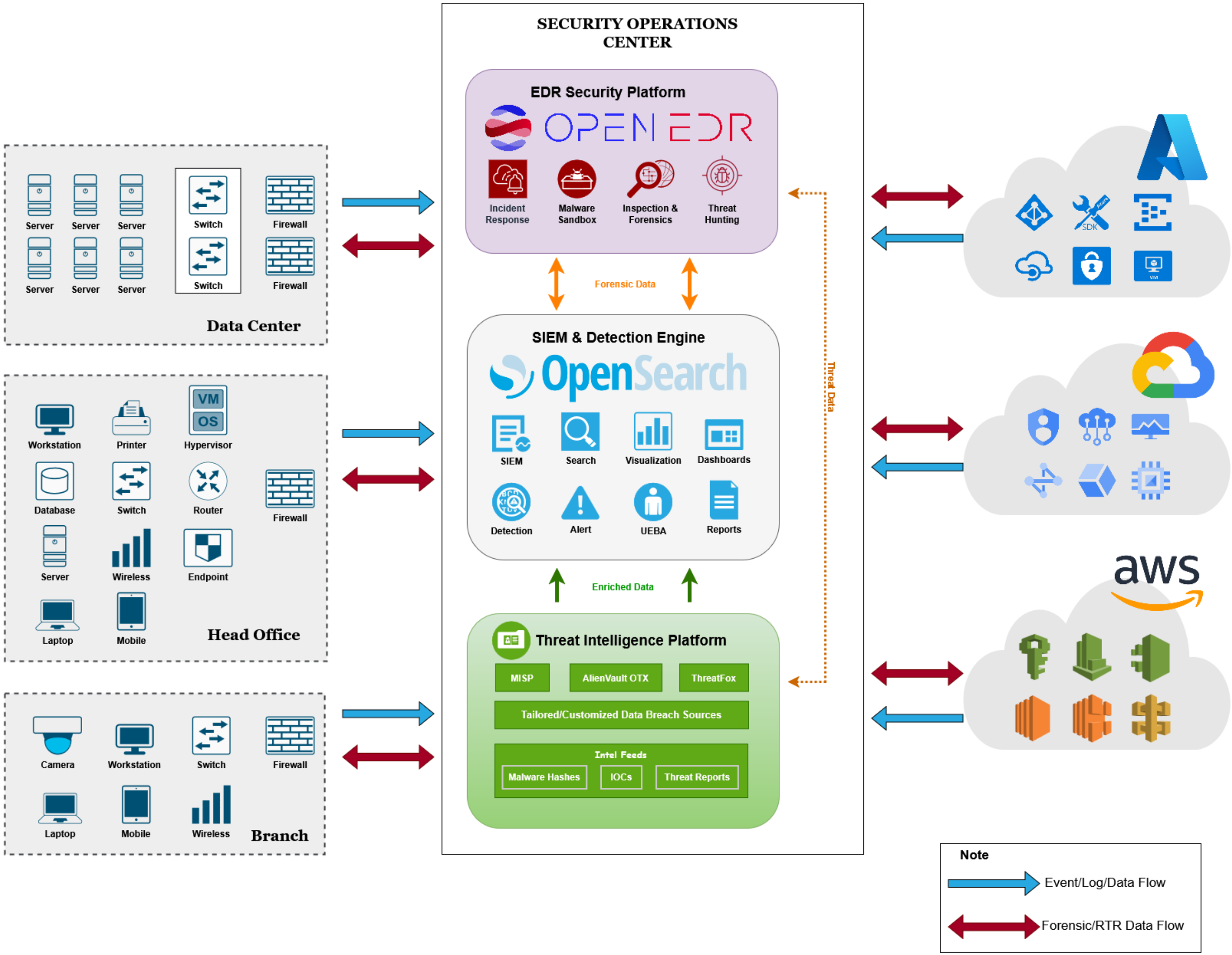
Network Engineer
SOC Engineer
Security Operations
Blue Teamer


SOC
VS

Overview of A SOC

A SOC is a centralized facility where information security professionals monitor, detect, analyze, and respond to cybersecurity incidents on a 24/7 basis

SOC Architecture



Core Functions

Continuous Monitoring

Real-time threat detection

...

Incident Response

Contain, eradicate, recover

...

Threat Intelligence

Staying ahead of emerging threats

...

Compliance Reporting

Meeting regulatory requirements

...

Forensic Analysis

Post-incident investigation

...

Technology Stack

SIEM

Log aggregation, correlation, alerting

SOAR

Automate repetitive tasks, orchestrate workflows

EDR/XDR

Endpoint visibility, behavioral analysis, response

Threat Intelligence Platforms

Endpoint visibility, behavioral analysis, response

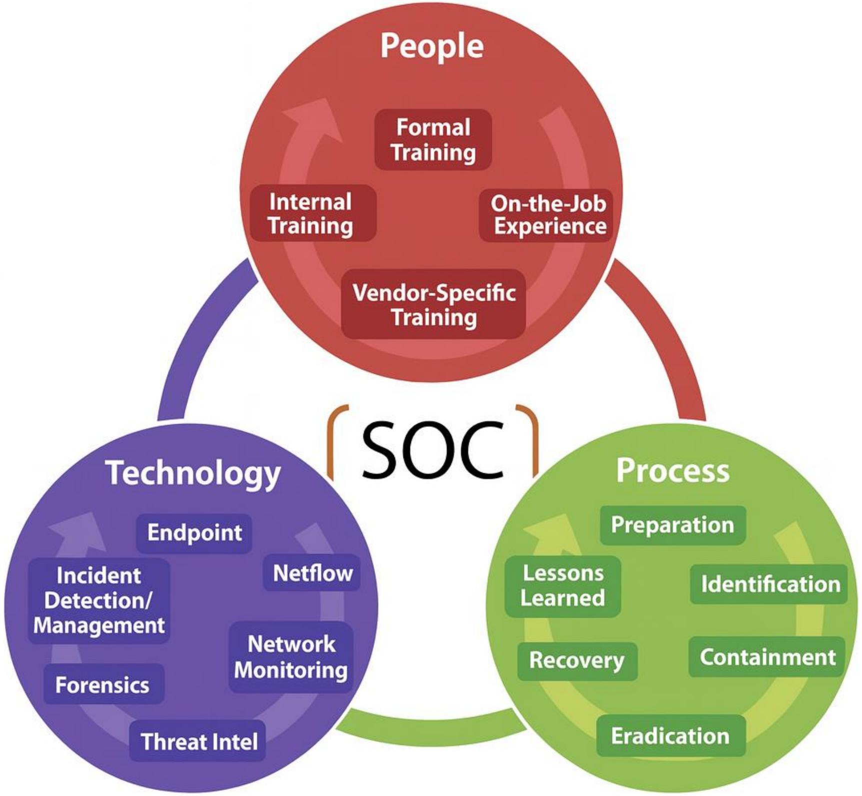
Network Security Monitoring

Network traffic analysis, lateral movement detection

Vulnerability Management

Asset discovery, vulnerability assessment, patch management

What Makes SOC Effective?



SOC Maturity Levels

01

Level 1 (Basic)

Reactive monitoring, basic alerting

02

Level 2 (Managed)

Proactive hunting, correlation rules

03

Level 3 (Optimized)

AI/ML integration, threat prediction

04

Level 4 (Adaptive)

Autonomous response, self-healing systems

Types of SOC Organizations

	In-House SOC	MSSP	Hybrid SOC	Virtual SOC
Pros	Full control, deep business context, customized processes	Cost-effective, expert staff, 24/7 coverage, latest tools	Balance of control and expertise, scalable, cost optimization	Cloud-native, highly scalable
Cons	High cost, staffing challenges, 24/7 coverage difficulty	Less customization, potential security concerns, dependency	Complexity in coordination, potential communication gaps	Coordination challenges, potential security risks
Best for	Large enterprises, regulated industries, high-security requirements	SMBs, organizations without security expertise	Mid-large enterprises wanting flexibility	Modern cloud-first organizations

A Day in the Life

SOC Engineer

- Design and optimize detection systems
- Automate manual processes
- Improve overall SOC efficiency

SOC Analyst

- Monitor and respond to alerts
- Follow established playbooks
- Investigate individual incidents

A Sample Alert

Alert

Alert: "Suspicious PowerShell execution detected"
- Source: 192.168.12.12
- Host OS: Windows
- Time: 2:47 AM
- User: nhu.pham@hoangnhu.lab
- Command: powershell.exe -enc <base64_string>

Initial Analysis Steps

1. Check if user was supposed to be online at 2:47 AM
2. Decode base64 string → reveals credential harvesting attempt
3. Check other endpoints for similar activity
4. Escalate to L2 for investigation

Challenges I Face Daily

Noisy Alerts

1000+ alerts per day in typical SOC
85-95% are false positives
Need to identify the 5-15% real threats quickly

Time Pressure

SLA requirements: P1 incidents within 15 minutes
Management pressure for quick resolution

High-Pressure Situations

Critical system compromises during business hours
Executive escalations and media attention
After-hours emergency response requirements

Challenges I Face Daily

Resource Constraints

Limited staff for 24/7 coverage
Budget constraints on new tools
Training time vs operational demands

Communication Skills

Explaining technical concepts to non-technical stakeholders
Writing clear incident reports
Presenting findings to management

Evolving Threat Landscape

New attack techniques daily
Zero-day vulnerabilities
Attackers adapting to detection methods

Skills

Technical Skills

- Networking, OS fundamentals
- Programing
- Scripting (Python, PowerShell)
- SIEM/Log analysis
- Basic malware analysis
- Cloud security basics

Soft Skills

- Critical thinking
- Communication
- Stress management
- Continuous learning mindset

Certifications

- CompTIA Security+
- CompTIA CySA+
- SANS
- GCIH (GIAC Certified Incident Handler)
- Vendor-specific Certificates
- OSDA SOC-200
- BTL1

Hand-on Labs

- Tryhackme
- Hackthebox
- Cyberdefenders

Career Paths

Vị trí	Kinh nghiệm yêu cầu	Mức lương/tháng (VND)	Mức lương/năm (USD)	Mô tả vai trò
SOC Manager	8-10 năm	60-90 triệu	$30,000-$45,000	Quản lý phòng thủ và ứng phó sự cố
IR Lead	6-8+ năm	55-80 triệu	$27,500-$40,000	Lãnh đạo ứng phó sự cố
SOC Lead	6-8+ năm	60-90 triệu	$30,000-$45,000	Quản lý trung tâm giám sát an ninh
Threat Intelligence Specialist	3-7 năm	45-70 triệu	$22,500-$35,000	Chuyên gia phân tích tình báo mối đe dọa
IR Specialist	3-7 năm	40-65 triệu	$20,000-$32,500	Chuyên gia ứng phó sự cố
SOC Engineer	2-6 năm	35-60 triệu	$17,500-$30,000	Chuyên viên triển khai SOC
SOC Analyst	2-4 năm	25-45 triệu	$12,500-$22,500	Chuyên viên giám sát và vận hành SOC

Tham khảo từ nhiều nguồn

Current Industry Trends

AI/ML Integration

- Behavioral analytics for anomaly detection
- Automated threat classification
- Predictive security modeling

Cloud-Native Security

- CASB (Cloud Access Security Brokers)
- CSPM (Cloud Security Posture Management)
- Container security monitoring

Zero Trust Architecture

- Never trust, always verify principle
- Micro-segmentation implementation
- Identity-centric security model

Threat Hunting Evolution

- Hypothesis-driven investigations
- MITRE ATT&CK framework adoption
- Collaborative purple team exercises

Advices

- How to get started (homelab, internship, blogs, ...)
- Certifications (Security+, CySA+, GCIH, ...)
- Networking

Thanks!

Do you have any questions?

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